

Human Vision – Topic 2

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1 Bayesian Decision Theory

In the last topic, we derived the closed-form solution for estimating velocity using the Lucas–Kanade (L-K) algorithm, which assumes small velocities and minimizes the Sum of Squared Differences (SSD) over an image patch. The solution reduces to solving the linear system $Av = b$, which relies on the matrix A being invertible:

$$A = \begin{bmatrix} \sum I_x^2 & \sum I_x I_y \\ \sum I_x I_y & \sum I_y^2 \end{bmatrix}.$$

What happens when A is not invertible? This occurs in two notable cases:

- **Constant image** ($I_x = I_y = 0$): for example, a blank patch of uniform intensity, in which case A is the zero matrix.
- **Gradients in only one direction**: for example, $I_y = 0$ along a perfectly vertical edge, so the second row and column collapse.

In these cases, the standard L-K algorithm is not well defined. Different algorithms adopt different strategies to cope with non-invertibility, and one of the simplest is regularization.

1.1 Regularized Lucas–Kanade

In the regularized L-K algorithm, we add a small constant $\lambda > 0$ to the diagonal of A , which guarantees invertibility:

$$A_{\text{reg}} = \begin{bmatrix} \sum I_x^2 + \lambda & \sum I_x I_y \\ \sum I_x I_y & \sum I_y^2 + \lambda \end{bmatrix}.$$

The resulting solution, however, does *not* strictly minimize the original SSD. If λ is chosen too large, the estimate is heavily biased and therefore inaccurate. This raises a natural question: *does adding λ to the diagonal actually make sense?* The justification, as we will see below, comes from Bayesian Decision Theory.

1.2 Bayesian Decision Theory — In Short

Suppose a thermometer shows 23.4°C. What is the temperature outside? The simple answer is 23.4°C — just trust the reading.

The Bayesian answer, however, depends on the type of thermometer and on historical data. If the historical average at this time of day is around 20.0°C, then prior experience suggests that the reading might be off, and we should fold this prior knowledge into our estimate.

Definition 1.1 Bayesian Formulation

Let X denote the true state of interest (the actual temperature) and let Y denote the observed measurement (the thermometer reading). Bayes' rule states

$$P(X | Y) = \frac{P(X) P(Y | X)}{P(Y)}.$$

1.3 Case of Gaussians

Suppose the prior and the likelihood are both Gaussian:

$$X \sim \mathcal{N}(\mu_P, \sigma_P^2), \quad Y | X \sim \mathcal{N}(X, \sigma^2).$$

Then the posterior is also Gaussian, $X | Y \sim \mathcal{N}(\mu_{x|y}, \sigma_{x|y}^2)$, with mean

$$\mu_{x|y} = \frac{\frac{1}{\sigma^2} Y + \frac{1}{\sigma_P^2} \mu_P}{\frac{1}{\sigma^2} + \frac{1}{\sigma_P^2}}.$$

Interpretation

The Bayesian posterior mean is a *weighted average* of the measurement Y and the prior mean μ_P , where each weight is the *inverse variance* (the precision) of the corresponding source. The larger the variance of a source, the less weight it receives; the more certain the source, the more it dominates the estimate.

Provided that our probabilistic model is correct on average, the Bayesian estimate is *optimal* in the sense that it minimizes the expected squared error.

Worked example. Returning to the thermometer scenario, we have $Y = 23.4$ and $\mu_P = 20.0$. The Bayesian estimate then depends on how reliable we believe the thermometer to be:

- If the thermometer is **very accurate** ($\sigma^2 \rightarrow 0$), the measurement dominates and the Bayes estimate tends to 23.4.
- If the thermometer is **very inaccurate** ($\sigma^2 \rightarrow \infty$), the prior dominates and the Bayes estimate tends to 20.0.
- For a **cheap thermometer** known to be noisy — i.e., σ^2 is large but finite — the estimate lies between the two, e.g., approximately 20.5.

1.4 A Philosophical Question

Suppose our cheap thermometer reads 23.4 and, given the prior, our Bayesian estimate is 20.5. We then take out a high-precision physics-lab thermometer and discover that the *true* temperature was in fact 23.4 — exactly what the cheap thermometer originally reported.

Is the Bayesian estimate of 20.5 then an *illusion*? In one sense, yes: on this particular occasion, the estimate disagrees with reality. In another sense, no: given only the noisy measurement and the prior, 20.5 was the optimal answer *in expectation*. This tension — between being correct on a single instance and being optimal on average — is the central idea behind Bayesian decision theory.

2 Combination Schemes for Motion

In Topic 1, we encountered three schemes for combining local motion measurements when an aperture is present:

1. **Vector Average (VA)** (VA): average the local normal-flow estimates.
2. **Intersection of Constraints (IOC)** (IOC): find the velocity consistent with all local constraint lines.
3. **Feature Tracking**: locate a distinctive feature (e.g. a corner or intersection) and track it over time.

Which method do humans actually use? VA and IOC frequently give different answers, and it turns out the perceptual answer is complicated: it depends on properties of the stimulus, and both VA and IOC outcomes have been reported in different experiments.

Consider a rhombus stimulus moving horizontally to the right. The IOC prediction places the velocity on the horizontal axis (the true motion), while the VA prediction yields a velocity that points diagonally downward, because the two visible edges have constraint lines whose unweighted average does not point horizontally:

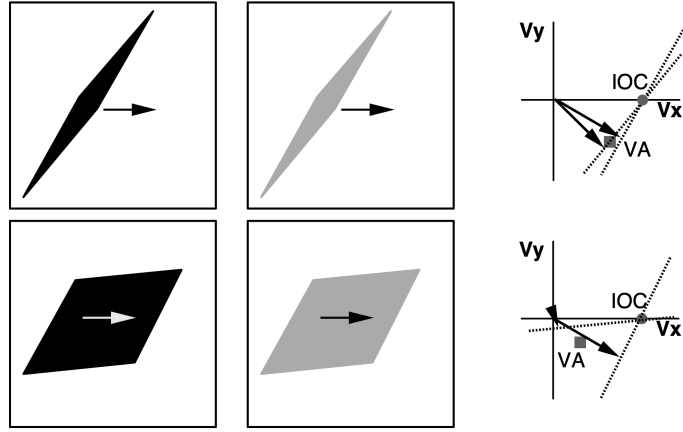


Figure 1: Two rhombus stimuli and their VA / IOC predictions in velocity space (Adelson & Movshon, 1982).

Several factors influence which percept dominates:

- **Shape of the rhombus.** When the rhombus is very thin, most observers report the diagonal direction (VA); when it is fat, observers report horizontal motion (IOC).
- **Background contrast.** Even with a thin rhombus, changing the background brightness can flip the reported percept.
- **Occlusion cues.** The presence of occluders also changes the report.

This patchwork of empirical findings motivates a principled framework that can *predict*, rather than just describe, when each percept arises.

2.1 Assumption 1 — Local Motion Measurements are Ambiguous and Noisy

The first assumption underlying the Bayesian motion model is that local motion measurements are inherently ambiguous and noisy:

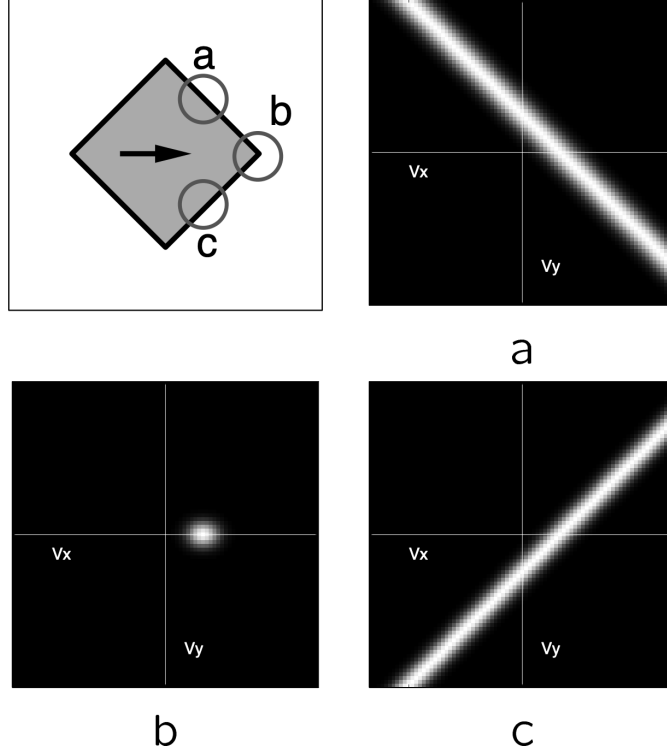


Figure 2: Local likelihoods at three sample locations (a, b, c) on a moving rhombus.

- At location **a**, the receptive field sees only an oriented edge with no endpoint, so we obtain a constraint line in velocity space. With added noise, the likelihood becomes a Gaussian-blurred ridge along that constraint line: in the figure, the brighter the pixel, the more probable that velocity.
- At location **b**, an endpoint (corner) is visible, so the velocity can in principle be pinned down to a single point. Due to noise, the localized likelihood still has non-zero probability over a small disc around the true velocity.
- At location **c**, we again see only an oriented edge — with a different orientation from **a** — giving a constraint-line likelihood with a different slope.

2.2 Assumption 2 — Velocities Tend to be Slow

The second assumption is that, in everyday experience, slow velocities are more common than fast ones (Wertheimer, 1912; Wallach, 1935). This assumption is equivalent to a “shortest path” heuristic for resolving motion correspondence problems, and reflects long-run statistics of motion in our visual environment.

To turn “slowness preference” into a Bayesian prior, we encode it as a Gaussian centred at zero in velocity space:

$$P(\vec{v}) \propto \exp\left(-\frac{\|\vec{v}\|^2}{2\sigma_P^2}\right).$$

2.3 Simple Bayesian Motion Estimator

We now combine the noisy likelihood from the constant-intensity (L-K) assumption with the slow-motion prior.

Likelihood. Defining the L-K cost

$$C(\vec{v}) = \sum_{x,y,t} w(x,y,t) (I_x v_x + I_y v_y + I_t)^2,$$

the local likelihood at a single location is

$$P(I(x,y) | \vec{v}) \propto \exp\left(-\frac{C(\vec{v})}{2\sigma_n^2}\right).$$

Prior. As above,

$$P(\vec{v}) \propto \exp\left(-\frac{\|\vec{v}\|^2}{2\sigma_P^2}\right).$$

Posterior. The *posterior* probability of a velocity given the observed image is, by Bayes' rule, proportional to likelihood times prior:

$$P(\vec{v} | I) \propto \left(\prod_{x,y} P(I(x,y) | \vec{v}) \right) P(\vec{v}).$$

This is the quantity we want: a probability distribution over velocities given everything we have observed. Picking the velocity that maximizes the posterior gives the maximum-a-posteriori (MAP) estimate.

Maximizing the posterior. Defining

$$M = \sum \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}, \quad b = - \sum \begin{bmatrix} I_x I_t \\ I_y I_t \end{bmatrix},$$

the MAP estimate satisfies

$$\vec{v} = \left(M + \frac{\sigma_n^2}{\sigma_P^2} I \right)^{-1} b,$$

where I here is the 2×2 identity matrix. There is a single free parameter, the ratio σ_n/σ_P , which can vary across experiments and subjects but is held fixed within a single experiment.

Connection back to Regularized L-K

This recovers the regularized Lucas–Kanade equation from Section 1, and identifies the previously-mysterious regularizer:

$$\lambda = \frac{\sigma_n^2}{\sigma_P^2}.$$

In words: λ is the ratio of measurement noise to prior uncertainty. Adding λ to the diagonal is therefore not an ad-hoc trick — it is exactly what falls out of doing optimal Bayesian inference under a slow-motion Gaussian prior.

2.4 Application: Predicting the Rhombus Percepts

The Bayesian estimator now lets us predict, rather than describe, when humans report VA versus IOC.

High-contrast thin rhombus. When contrast is high, the likelihood is sharply peaked along each constraint line (small σ_n^2). Multiplying two narrow constraint likelihoods gives a posterior dominated by their intersection, and the slow prior has little effect. The MAP estimate coincides with the IOC solution.

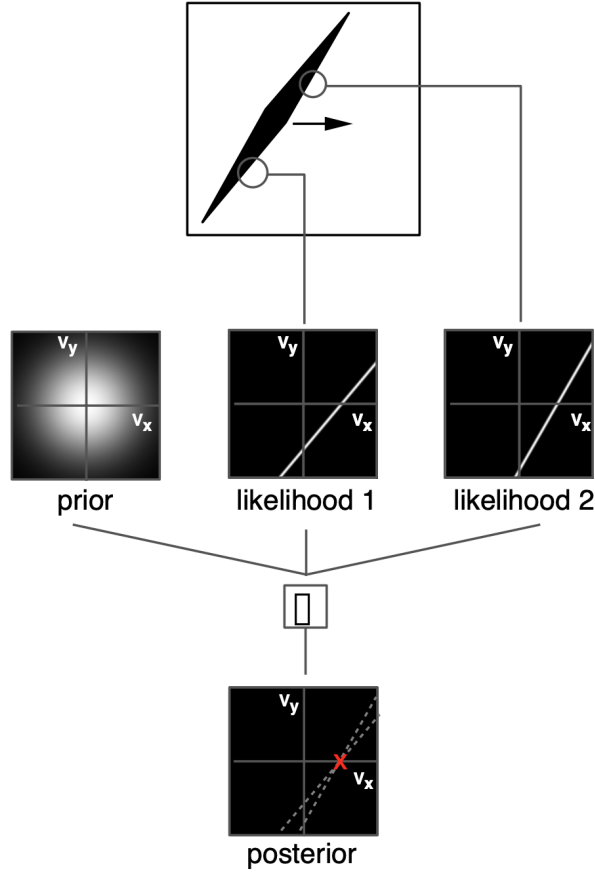


Figure 3: High-contrast rhombus: narrow constraint likelihoods produce an IOC-like percept.

Low-contrast thin rhombus. When contrast is reduced (e.g. by changing the background), the likelihood ridges become broad and uncertain. The slow prior now meaningfully pulls the posterior toward the origin, and the MAP estimate lies diagonally downward — close to the VA solution.

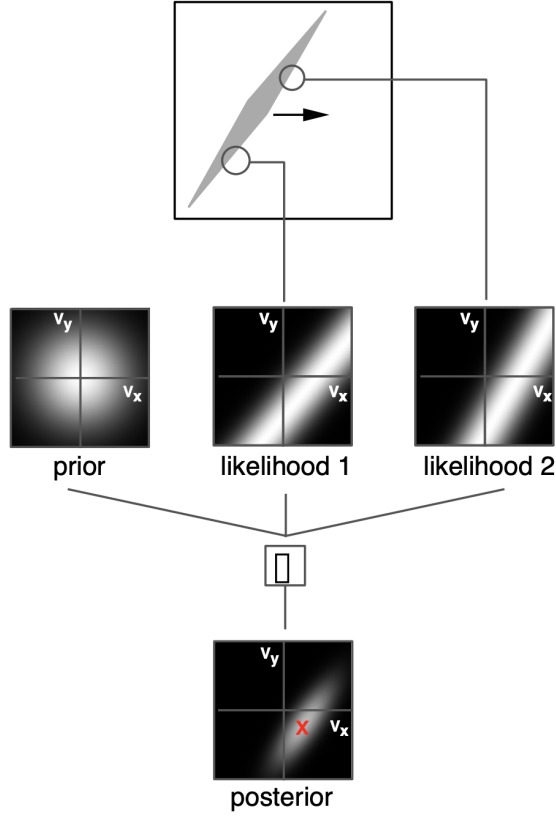


Figure 4: Reduced-contrast rhombus: broad likelihoods let the slow prior dominate, producing a VA-like percept.

Fat rhombus. For a fat rhombus, the visible edges have orientations closer to vertical, so their constraint lines intersect on the horizontal axis. The IOC and the slow prior agree on horizontal motion, and the MAP estimate is horizontal.

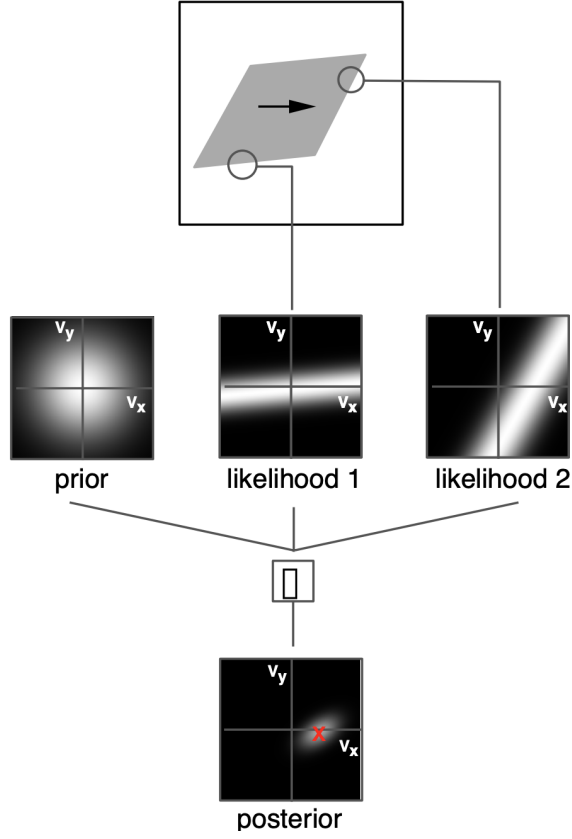


Figure 5: Fat rhombus: constraint geometry favours horizontal IOC motion.

Across many displays. Surveying many such stimuli, the Bayesian estimate tracks human reports more closely than either VA or IOC alone: IOC reports cluster near horizontal, VA reports cluster near diagonal, and the Bayesian (and human) percept lies in between, shifting smoothly with contrast and shape.

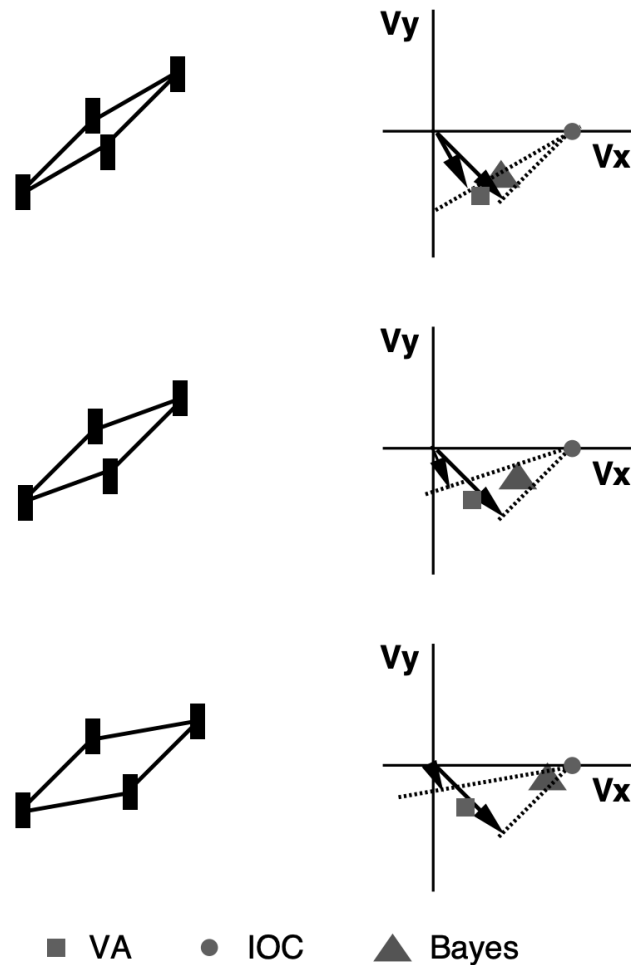


Figure 6: Across stimuli: IOC near horizontal, VA near diagonal, Bayesian (and human) in between.

Optimal but not always accurate

The Bayesian estimator is optimal in the sense of minimizing expected error under its assumptions, but optimality does not imply that perceived motion is always faithful to the physical stimulus. A well-known applied case is driving in fog: low contrast makes oncoming visual motion appear slower than it really is, biasing speed perception toward the slow prior. Drivers therefore tend to drive too fast in fog — exactly as the Bayesian model predicts (Snowden, Stimpson, & Ruddle, 1998).

3 Quiz

Question 1. Two plaids are physically moving with the same velocity. The first is high-contrast (white grid on a black background); the second is low-contrast (light-grey grid on a grey background). Which appears to move faster?



Figure 7: Quiz Q1: two plaids moving at the same physical velocity, one high-contrast and one low-contrast.

Answer. The high-contrast plaid (the one on the dark background) appears to move *faster*.

Why. This is the contrast–speed effect of Thompson et al. (1996): at the same physical speed, lower-contrast stimuli are perceived as slower. The Bayesian motion model explains this directly. Lower contrast produces broader, more uncertain likelihood ridges (large effective σ_n^2), so the slow prior pulls the MAP estimate further toward zero. Higher contrast produces sharper likelihoods, so the prior has less influence and the MAP estimate stays closer to the true (faster) velocity.

Question 2. An ultrasound measurement gives a fetus’ head size of 10.0 cm. The measurement is known to be noisy. The prior on head size at this gestational stage has mean $\mu_P = 15$ and standard deviation $\sigma_P = 3$. Will the Bayesian estimate be *below*, *above*, or *equal to* 10.0?

Answer. The Bayesian estimate will be *above* 10.0.

Why. Recall the Gaussian–Gaussian posterior mean from Section 1.3:

$$\mu_{x|y} = \frac{\frac{1}{\sigma^2} Y + \frac{1}{\sigma_P^2} \mu_P}{\frac{1}{\sigma^2} + \frac{1}{\sigma_P^2}}.$$

With $Y = 10$ and $\mu_P = 15$, the estimate is a convex combination of 10 and 15 with strictly positive weights on both. As long as the ultrasound noise σ is finite and non-zero (i.e., the measurement is noisy at all but not infinitely so), the estimate satisfies $10 < \mu_{x|y} < 15$. The estimate is “pulled toward the prior” because the prior has non-zero precision $1/\sigma_P^2 = 1/9$.